

Infection Control in The Surgical Practice

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Health care professionals must learn and practice protocols that limit the spread of contagions in the patient care setting. This is especially true for dentists performing surgery for two reasons: First, to perform surgery, the dentist typically violates an epithelial surface, the most important barrier against infection. Second, during most oral surgical procedures, the dentist, assistants, and equipment become contaminated with the patient's blood and saliva.

Communicable Pathogenic Organisms

In the case of oral surgery, the surgeons are fighting virulent bacteria, mycobacteria, fungi, and viruses. The strengths of these organisms are the various means that they use to prevent their elimination, whereas their weaknesses are their susceptibilities to chemical, biologic, and physical agents. By understanding the “enemy,” the dentist can make rational decisions about infection control.

Bacteria

Upper Respiratory Tract Flora

Normal oral flora contains the microorganisms usually present in saliva and on the surfaces of oral tissues in healthy, immunocompetent individuals who have not been exposed to agents that alter the composition of oral organisms.

Normal Microbiologic Flora

Region	Bacteria
Oral cavity	Aerobic gram-positive organisms, primarily <i>Streptococcus</i> spp., <i>Actinomyces</i> spp. Anaerobic bacteria, including <i>Prevotella melaninogenica</i> <i>Candida</i> spp.
Nasal cavity	Aerobic gram-positive organisms, primarily <i>Streptococcus</i> spp. In children, <i>Haemophilus influenzae</i> frequently present In adults, <i>Staphylococcus aureus</i> frequently present
Facial skin	<i>Staphylococcus</i> spp., primarily <i>S. epidermidis</i> , occasionally <i>S. aureus</i> <i>Corynebacterium diphtheriae</i> <i>Propionibacterium acnes</i>
All areas below clavicles, including hands	<i>S. epidermidis</i> <i>C. diphtheriae</i> Gram-negative aerobes such as <i>Escherichia coli</i> , <i>Klebsiella</i> spp., and <i>Proteus</i> spp. Anaerobic enteric organisms, including <i>Bacteroides fragilis</i>

The total number of oral organisms is held in check by the following four main processes:

- (1) Rapid epithelial turnover with desquamation.
- (2) Host immunologic factors such as salivary immunoglobulin A.
- (3) Dilution by salivary flow.
- (4) Competition between oral organisms for available nutrients and attachment sites.

The flora of the nose and paranasal sinuses consists primarily of gram-positive aerobic streptococci and anaerobes. In addition, many children harbor *Haemophilus influenzae* bacteria in these areas, and many adults have *Staphylococcus aureus* as a part of their transient or resident nasal and paranasal sinus flora.

The skin of the maxillofacial region has surprisingly few resident organisms in its normal flora. The bacteria *Staphylococcus epidermidis* and *Corynebacterium diphtheriae* are the predominant species present. *Propionibacterium acnes* is found in pores and hair follicles, and many individuals carry *S. aureus*, spread from the nose on the facial skin

Viral Organisms

Viruses are ubiquitous in the environment, but, fortunately, only a few pose a serious threat to the patient and the surgical team. The viral organisms that cause the most difficulty are the hepatitis B and C viruses and human immunodeficiency virus (HIV). These viruses have differences in their susceptibility to inactivation that are important to understand when attempting to prevent their spread.

Hepatitis Viruses

Hepatitis A, B, C, and D viruses are responsible for most infectious hepatic diseases.

Hepatitis A is spread primarily by contact with the feces of infected individuals.

Hepatitis C virus may spread through contaminated feces or by contaminated blood.

Hepatitis B and D viruses are spread by contact with any human secretion. In addition to preventing patient-to-patient spread, the dentist and the staff also need to take precautions to protect themselves from contamination because in several instances, dentists have been the primary source of a hepatitis B epidemic

Methods Designed to Limit the Spread of Hepatitis Viruses

From Infected Patient to Other Patients

- Use disposable materials.
- Disinfect surfaces.
 - A. With halogen compounds:
 1. Iodophors
 2. Hypochlorite (bleach)
 - B. With aldehydes:
 1. Formaldehyde
 2. Glutaraldehyde
- Sterilize reusable instruments.
 - A. With heat
 - B. With ethylene oxide gas
- Use disposable materials.

From Infected Patient to Dental Staff

- Learn to recognize individuals likely to be carriers.
- Use barrier techniques (e.g., gloves, facemask, and eye protection) during surgery, when handling contaminated objects, and during cleanup.
- Promptly dispose of sharp objects into well-labeled protective containers.
- Dispose of needles immediately after use or resheathe in-use instruments.
- Use an instrument to place a scalpel blade on or take one off of a blade handle.
- Ensure hepatitis B vaccination of dental staff.

Universal precautions typically include having all doctors and staff who come in contact with patient blood or secretions, whether directly or in aerosol form, wear barrier devices, including a face-mask, hair coverage, eye protection, and gloves. Universal precaution procedures go on to include decontaminating or disposing of all surfaces that are exposed to patient blood, tissue, and secretions. Finally, universal precautions mandate avoidance of touching, and thereby contaminating, surfaces (e.g., the dental record, computer keyboard, uncovered light handles, and telephone) with contaminated gloves or instruments.

Human Immunodeficiency Virus

Because of its relative inability to survive outside the host organism, HIV (the causative agent of acquired immunodeficiency syndrome), acts in a fashion similar to other agents of sexually transmitted diseases. That is, transfer of the virions from one individual to another requires direct contact between virus-laden blood or secretions from the infected host organism and a mucosal surface or epithelial wound of the potential host. Evidence has shown that HIV loses its infectivity once desiccated.

Mycobacterial Organisms

The only mycobacterial organism of significance to most dentists is *Mycobacterium tuberculosis*. Although tuberculosis (TB) is an uncommon disease in the United States and Canada, the frequent movement of persons between countries, including those where TB is common, continues to spread *M. tuberculosis* organisms worldwide. TB is transmitted primarily through exhaled aerosols that carry *M. tuberculosis* bacilli. However, transmission can also occur via inadequately sterilized instruments because, although *M. tuberculosis* organisms do not form spores, they are highly resistant to desiccation and to most chemical disinfectants.

Aseptic Techniques

Antiseptic and disinfectant are terms that are often misused. Both refer to substances that can prevent the multiplication of organisms capable of causing infection. The difference is that antiseptics are applied to living tissue, whereas disinfectants are designed only for use on inanimate objects.

Sterility is the freedom from viable forms of microorganisms. Sterility represents an absolute state; there are no degrees of sterility. Sanitization is the reduction of the number of viable microorganisms to levels judged safe by public health standards. Sanitization should not be confused with sterilization.

Dental office sterility levels

- critical,
- semi-critical,
- and noncritical. Following the Centers for Disease Control and Prevention guidelines, these levels dictate how tools are cleaned, disinfected, or heat-sterilized between patients.

Critical Items

- **Risk level:** Highest risk; these tools penetrate soft tissue, bone, or enter into the bloodstream.
- **Examples:** Surgical scalpels, periodontal scalers, surgical forceps, and bone chisels.
- **Requirement:** Must always be heat-sterilized after every use or safely discarded if single-use.

Semi-critical items

- **Risk level:** Medium risk; these items touch mucous membranes or non-intact skin, but do not break through bone or tissue.
- **Examples:** Dental mouth mirrors, reusable impression trays, and air-water syringes.
- **Requirement:** Must be heat-sterilized if heat-tolerant; heat-sensitive items require at least high-level disinfection, though heat sterilization is mandatory for dental handpieces.

Noncritical Items

- **Risk level:** Lowest risk; these surfaces or objects touch only intact skin and not mucous membranes.
- **Examples:** Dental chair surfaces, countertops, light handles, and overhead equipment.
- **Requirement:** Cleaned and barrier-protected using low- to intermediate-level EPA-registered hospital disinfectants between patients.

Techniques of Instrument Sterilization

Any means of instrument sterilization to be used in office-based dental and surgical care must be reliable, practical, and safe for the instruments. The three methods generally available for instrument sterilization are dry heat, moist heat, and ethylene oxide gas.

Box 5.2

General Methods of Reducing the Number of Viable Organisms From a Surface

Physical

- Heat
- Mechanical dislodgment
- Radiation

Chemical

- Antiseptics
- Disinfectants
- Ethylene oxide gas

Sterilization With Heat

Dry Heat

Dry heat is a method of sterilization that can be provided in most dental offices because of the ease of production. Dry heat is most commonly used to sterilize glassware and bulky items that can withstand heat but are susceptible to rust. The time necessary for dry heat sterilization limits its practicality in the ambulatory setting because it lengthens the turnover time and forces the dentist to have many duplicate instruments. The advantages of dry heat are the relative ease of use and the unlikelihood of damaging heat-resistant instruments.

Guidelines for Dry Heat and Steam Sterilization ^a

Temperature	Duration of Treatment or Exposure ^a
Dry Heat	
121°C (250°F)	6–12?h
140°C (285°F)	3?h
150°C (300°F)	2.5?h
160°C (320°F)	2?h
170°C (340°F)	1?h
Steam	
116°C (240°F)	60?min
118°C (245°F)	36?min
121°C (250°F)	24?min
125°C (257°F)	16?min
132°C (270°F)	4?min
138°C (280°F)	1.5?min

Moist Heat

Moist heat sterilization is more efficient than dry heat sterilization because it is effective at much lower temperatures and requires less time. The container usually used for providing steam under pressure is known as an autoclave. The autoclave works by creating steam and then, through a series of valves, increases the pressure so that the steam becomes super-heated. Instruments placed into an autoclave should be packaged to allow the free flow of steam around the instruments, such as by placing them in sterilization pouches or wrapping them in cotton cloth.



Comparison of Dry Heat Sterilization vs. Moist Heat Sterilization Techniques

	Dry Heat	Moist Heat
Principal antimicrobial effect	Oxidizes cell proteins	Denatures cell proteins
Time necessary to achieve sterilization	Long	Short
Equipment complexity and cost	Low	High
Tendency to dull or rust instruments	Low	High
Availability of equipment sized for office use	Good	Good

Sterilization With Gas

Certain gases exert a lethal action on bacteria by destroying enzymes and other vital biochemical structures. Of the several gases available for sterilization, ethylene oxide is the most commonly used.

Techniques of Instrument Disinfection

Many dental instruments cannot withstand the temperatures required for heat sterilization, therefore if sterilization with gas is not available and absolute sterility is not required, chemical disinfection can be performed. Chemical agents with potential disinfectant capabilities have been classified as being high, intermediate, or low in biocidal activity.

Classification System for the Biocidal Effects of Chemical Disinfectants

Level of Biocidal Activity ^a	Vegetative Bacteria	Lipid Viruses	Nonlipid Viruses	Tubercle Bacilli	Bacterial Spores
Low	+	+	—	—	—
Intermediate	+	+	+	+	
High	+	+	+	+	+

Biocidal Activity of Various Chemical Disinfectants

			ACTIVITY LEVEL ^a	
Generic	Brand Names	Exposure Time	Intermediate	High
Formaldehyde 3%				
8% or		≥30?min	+	
8% in 70% alcohol		10?h		
Glutaraldehyde 2% with nonionic ethoxylates of linear alcohol	Wavicide, Sterall			
Room temperature		≥10?min	+	
40°C–45°C		4?h		+
60°C		4?h		+
Glutaraldehyde 2% alkaline with phenolics buffer	Sporicidin			
Diluted 1 : 6		≥10?min	+	
Full strength		7?h		+
Glutaraldehyde 2% alkaline	Cidex, Procide, Glutarex, Omnicide	≥10?min	+	
		10?h		+
1% Chlorine compound, diluted 1 : 5	Clorox	≥30?min	+	
O-phenylphenol 9% plus O-benzyl-p-chlorophenol 1%, diluted 1 : 32	Omni II	≥10?min	+	
Iodophors 1% iodine	Betadine, Isodine	≥30?min	+	

Methods of Sterilization or Disinfection of Selected Dental Instruments

Items	Steam Autoclave (15–30 Min Required per Cycle)	CHEMICAL DISINFECTION	
		Dry Heat Oven (1–1.5 Hours Required per Cycle)	Sterilization ^a
Stainless instruments (loose) restorative burs	++	++	–
Instruments in packs	++	+ (Small packs)	–
Instrument tray setups, surgical or restorative	+ (Size limit)	++	–
Rust-prone instruments	(Only when coated with chemical protectant)	++	
Handpiece (autoclave)	++	–	–
Handpiece (nonautoclave)		–	± (Iodophor disinfectant)
Angle attachment ^b	+	+	–
Rubber items	++	–	–
Rag wheels	++	+	–
Removable prosthetics ^c	–	–	
Heat-resistant plastic evacuators	++	+	–

Maintenance of Sterility

Disposable Materials

Materials and drugs used during oral and maxillofacial surgery—such as sutures, local anesthetics, scalpel blades, and syringes with needles—are sterilized by the manufacturer with a variety of techniques, including use of gases, autoclaving, filtration, and irradiation. To maintain sterility, only the dentist must properly remove the material or drug from its container. Most surgical supplies are double wrapped (the only common exception is scalpel blades).



Surgical Field Maintenance

Anesthetic equipment used to deliver gases such as oxygen or nitrous oxide may also spread infection from patient to patient. Plastic nasal cannulas are designed to be discarded after one use. Nasal masks and the tubing leading to the mask from the source of the gases are available in disposable form or can be covered with disposable sleeves.

Surgical wound Classification

Table 1

Surgical Wound Classification Grades (I–IV) as Defined by the CDC

CDC Surgical Wound Classification Definitions

Class I/Clean: An uninfected operative wound in which no inflammation is encountered, and the respiratory, alimentary, genital, or uninfected urinary tract is not entered. In addition, clean wounds are primarily closed and, if necessary, drained with closed drainage. Operative incisional wounds that follow no penetrating (blunt) trauma should be included in this category if they meet the criteria.

Class II/Clean-Contaminated: An operative wound in which the respiratory, alimentary, genital, or urinary tracts are entered under controlled conditions and without unusual contamination. Specifically, operations involving the biliary tract, appendix, vagina, and oropharynx are included in this category, provided no evidence of infection or major break in a sterile technique is encountered.

Class III/Contaminated: Open, fresh, accidental wounds. In addition, operations with major breaks in a sterile technique (eg, open cardiac massage) or gross spillage from the gastrointestinal tract, and incisions in which acute or no purulent inflammation is encountered are included in this category.

Class IV/Dirty-Infected: Old traumatic wounds with retained devitalized tissue and those that involve existing clinical infection or perforated viscera. This definition suggests that the organisms causing postoperative infection were present in the operative field before the operation.

CDC = Centers for Disease Control and Prevention.

Surgical Staff Preparation

Hand and Arm Preparation

Antiseptics are used to prepare the surgical team's hands and arms before gloves are donned and are also used to disinfect the surgical site. Because antiseptics are used on living tissue, they are designed to have low tissue toxicity while maintaining disinfecting properties. The three antiseptics most commonly used in dentistry are (1) iodophors, (2) chlorhexidine, and (3) hexachlorophene.

Clean Technique

The clean technique is generally used for office-based surgery that does not specifically require a sterile technique. Office oral surgical procedures that call for a sterile technique include any surgery in which skin is incised. The clean technique is designed as much to protect the dental staff and other patients from a particular patient as it is to protect the patient from pathogens that the dental staff may harbor.

When using a clean technique, the dental staff may wear clean street clothing covered by long-sleeved laboratory coats



Dentists should wear sterile gloves whenever they are providing invasive dental care. When the clean technique is used, hands may be washed with antiseptic soap and dried on a disposable towel before gloving. Gloves should be sterile and put on using an appropriate technique to maintain sterility of the external surfaces.





In general, eye protection should be worn when blood or saliva are dispersed, such as when high-speed cutting equipment is used. A facemask and hair coverage should be used whenever aerosols are created or a surgical wound is to be made. In most cases, it is not absolutely necessary to prepare the operative site when using the clean technique. However, when surgery in the oral cavity is performed, perioral skin may be decontaminated with the same solutions used to scrub the hands, and the oral cavity may be prepared by brushing or rinsing with chlorhexidine gluconate (0.12%) or an alcohol-based mouthwash.

These procedures reduce the amount of skin or oral mucosal contamination of the wound and decrease the microbial load of any aerosols made while using high-speed drills in the mouth. The dentist may desire to drape the patient to protect the patient's clothes, to keep objects from accidentally entering the patient's eyes, and to decrease suture contamination should it fall across an uncovered, unprepared part of the patient's body.

During an oral surgical procedure, only sterile water or sterile saline solution should be used to irrigate open wounds. A disposable injection syringe, a reusable bulb syringe, or an irrigation pump connected to a bag of intravenous solution can be used to deliver irrigation. Reservoirs feeding irrigation lines to handpieces are also available and can be filled with sterile irrigation fluids.

Sterile Technique

The sterile technique is used for office-based surgery when skin incisions are made or when surgery is performed in an operating room. (A clean wound is made through intact skin that has been treated with an antiseptic.) The purpose of sterile technique is to minimize the number of organisms that enter wounds created by the surgeon. The technique requires strict attention to detail and cooperation among the members of the surgical team.

The surgical hand and arm scrub is another means of lessening the chance of contaminating a patient's wound. Although sterile gloves are worn, gloves can be torn (especially when using high-speed drills or working around jaw fixation wires), thereby exposing the surgeon's skin. By proper scrubbing with antiseptic solutions, the surface bacterial level of the hands and arms is greatly reduced.

Most hospitals have a surgical scrub protocol that must be followed when performing surgery in those institutions. Although several acceptable methods can be used, standard to most techniques is the use of an antiseptic soap solution, a moderately stiff brush, and a fingernail cleaner.

Hands and forearms are washed in a scrub sink, and hands are kept above the level of the elbows after washing until the hands and arms are dried. A copious amount of antiseptic soap is applied to the hands and arms from either wall dispensers or antiseptic-impregnated scrub brushes. The antiseptic soap is allowed to remain on the arms, while any dirt is removed from underneath each fingernail tip using a sharp-tipped fingernail cleaner.

Postsurgical Asepsis

Wounds Management

A few principles of postsurgical care are useful to prevent the spread of pathogens. Wounds should be inspected or dressed by hands that are covered with fresh, clean gloves. When several patients are waiting, those without infectious problems should be seen first, and those with problems such as a draining abscess should be seen afterward.

Thank you